

Integrative bioinformatics and transcriptomic analysis identifies biomarkers in Polycystic Ovary Syndrome through machine learning approach

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ABSTRACT

Polycystic Ovary Syndrome (PCOS) is a common endocrine condition that affects women of reproductive age. The study used high-throughput sequencing to profile gene expression in PCOS and control samples. The sequenced reads were quality assessed and aligned to the human reference genome hg38 using FastQC and the Hisat2 aligner. The R software "DESeq2" identified 1193 differentially expressed genes (DEGs) in SRP224633 and 82 DEGs in SRP353681. Notably, 8 DEGs were shared by the two datasets, and a total of 1267 DEGs were found. GO analysis revealed that DEGs in SRP224633 were enriched in biological processes related to immune response and cell activation, whereas DEGs in SRP353681 were associated with response to external stimuli and immune processes. Pathway analysis highlighted the involvement of chemokine signaling receptor, and cytokine-cytokine receptor interaction pathways in both datasets. The STRING database was used to evaluate protein-protein interaction (PPI) networks. Hub genes like IL1B, PTPRC, ITGAM, TYROBP, ITGB2, FCGR3A, CCR7, SYK, TLR2, FCGR3B were identified as crucial nodes using Cytohubba plugins in Cytoscape, highlighting their potential role in PCOS pathogenesis. Regulatory networks discovered miRNAs such as has-mir-34a-5p, hsa-miR-26a-5p, has-let-7b-5p and transcription factors such as SP1, RUNX1, ER as possible regulators of target genes implicated in PCOS pathogenesis. LASSO regression and ROC analysis indicated certain hub genes, such as IL1B, PTPRC, ITGB2, FCGR3A, CCR7, FCGR3B as relevant biomarkers due to their high coefficients, emphasizing their significance in PCOS. CCR7, FCGR3A, FCGR3B, and ITGB2 have been discovered as novel genes with significant potential as PCOS biomarkers. Our research adds to a deeper knowledge of PCOS at the molecular level by offering an extensive understanding of the etiological reasons and molecular mechanisms. The discovered DEGs, pathways, and regulatory networks and novel biomarker are promising targets for future studies and treatments in PCOS management.

1. Introduction

The combination of bioinformatics with next-generation sequencing (NGS) technology, also known as high throughput sequencing, has rendered the ability to identify, analyze and detect human disease with tremendous accuracy [1]. In particular, RNA-sequencing (RNA-seq) techniques offer an abundance of research information at a high resolution on the gene expression patterns of various species under different conditions [2–4]. The concept of differently expressed genes (DEGs), or genes that their expression levels have been found to be extensively differentially expressed between two or more conditions [5,6]. Polycystic ovarian syndrome (PCOS) is a common endocrine disorder that

mostly affects women, especially those of reproductive age. According to estimates, PCOS affects 5–10 % of people worldwide. Adolescents with PCOS frequently exhibit hyperandrogenism and irregular menstruation [7]. In 2012, the National Institutes of Health sponsored the PCOS Workshop, which divided PCOS into four different phenotypes. The following phenotypes are classified: phenotype A, which includes polycystic ovary morphology, ovulatory dysfunction and hyperandrogenism; phenotype B, which includes, ovulatory dysfunction and hyperandrogenism; phenotype C, which includes polycystic ovary morphological characteristics and hyperandrogenism; and phenotype D, which includes both polycystic ovary morphology and ovulatory dysfunction [8,9]. In order to find novel diagnostic indicators, prognosis

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markers and treatment targets for PCOS, it is crucial to understand the etiological causes, molecular processes and pathways. Furthermore, high-throughput RNA sequencing has been used to investigate gene expression profiling in PCOS, leading to the identification of several key genes and diagnostic biomarkers for this syndrome. This investigation has included profiling numerous differentially expressed genes (DEGs) linked to various pathways, biological processes, or molecular functions [10]. The focus of differential gene expression tools is to identify the genes that differ statistically significant and to provide information about each gene's expression level and pairwise magnitude of difference. These tools are based on quantifications of expressed genes obtained from computational analyses of raw RNA-seq reads [11] e.g., mapping [12] and assembly [13–15].

Bioinformatics and Microarray-based analyses have already been employed to study gene expression alterations in PCOS [16,17]. Our research used RNA-Seq data, which provides greater accuracy, sensitivity and resolution compared to the traditional microarray technologies. This allowed us to perform comprehensive transcriptomic analysis and detect subtle changes in gene expression between PCOS and control populations, providing robust and reliable insights. We have uniquely integrated widely used bioinformatic techniques such as GO and KEGG pathway analyses with LASSO regression for feature selection, enabling the identification of DEGs with high predictive potential as biomarkers for PCOS. This combination of approaches has not been fully explored in previous studies, particularly with NGS data. In addition, we identified several candidate genes as potential biomarkers for PCOS that have not been previously reported, providing new insights into its molecular mechanisms.

In this current study, RNA-Seq data for PCOS are used to conduct a thorough analysis of DEGs. The process comprised obtaining raw data from the National Center for Biotechnology Information (NCBI-SRA) and the quality of data was assessed through FastQC. The reads were then trimmed and aligned. In order to identify the expression patterns associated with PCOS, an extensive examination of the DEGs was conducted in R as described in Fig. 1. Major DEGs were analyzed for pathway and functional enrichment, as well as protein-protein interaction (PPI) network to depict key genes in cytoscape. The miRNA-TF and target gene regulatory network depicts the relationships between transcription factors (TFs), miRNAs, and their key targets in PCOS. The use of LASSO Poisson regression analysis on the identified key genes (*IL1B*, *PTPRC*, *ITGAM*, *TYROBP*, *ITGB2*, *FCGR3A*, *CCR7*, *SYK*, *TLR2* and *FCGR3B*) revealed novel gene findings about their function in PCOS. The objective of this study was to use bioinformatics analysis to discover key genes and pathways associated with PCOS. Subsequently, the research aimed to investigate the molecular processes underlying PCOS and classify novel treatment biomarkers for PCOS diagnosis. These studies aim to broaden the comprehension of PCOS etiology at the molecular level.

2. Materials and methods

2.1. RNA-SEQ data retrieval

High-throughput sequencing datasets of paired-end sequence such as SRP224633 [18] and SRP353681 [19] were retrieved from NCBI-Sequence Read Archive (SRA), an open-access repository of next-generation sequencing, to identify differentially expressed genes (DEGs) between PCOS and control samples. It consist of 17 granulosa cell samples: 9 PCOS and 8 control samples. To ensure the adequacy of the sample size, we performed a statistical power analysis using the SSizer tool [20]. The sequence read archive format of the paired-end sequence data was downloaded using the SRA toolkit.

2.2. Pre-processing of the data

Adaptor or primer contamination can result in low-quality reads in

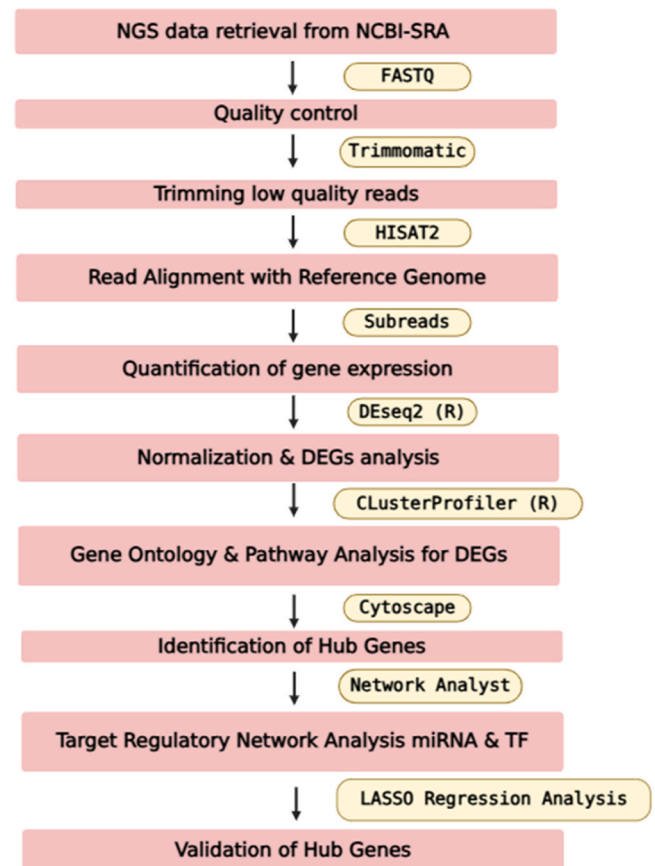


Fig. 1. Schematic representation of steps involved for the identification of hub gene in PCOS.

raw data reads generated by contemporary high-throughput sequencers. Therefore, quality control tests were carried out using the FastQC tool in order to ensure the base quality and eliminate noise from the data [21].

2.3. Alignment of the reads

The pre-processed Fastq files were mapped using HISAT2. RNA-Seq reads are frequently aligned to a reference genome hg38 using HISAT2 in order to measure the levels of gene expression [22]. The outcome of the alignment step is frequently in SAM format, which is text-based. SAM files are converted to binary format (BAM) using tools like Samtools to reduce file size and improve efficiency [23]. A GTF (Gene Transfer Format) file is widely used in combination with HISAT2 to give annotation information for genomic characteristics. Subreads assign reads to annotate genomic features and generate a count matrix representing the total number of reads mapped to each gene in each sample [24].

2.4. Identification of DEGs

The data generated from FeatureCounts was utilized in the DESeq2 package in the R program (<https://www.r-project.org>) to identify genes that are differentially expressed. DEGs between the PCOS group and the normal group were defined as genes with $P_{adj} < 0.05$ and $|\log_{2}FC| \geq 1$. DESeq2 implements the Benjamini-Hochberg false discovery rate (FDR) as the default approach for multiple test correction [25].

2.5. GO and pathway enrichment of DEGs

The R package "org.hs.eg.db" was utilized to change the Ensembl

gene ID of DEGs into gene names. Cluster Profiler and enrichPATHWAY package in R was used for enrichment analysis, with $Padj < 0.05$ indicating statistical significance [26]. Three terminologies have been used to represent gene functions in Gene Ontology [GO], molecular function (MF), cellular component (CC) and biological processes (BP). Kyoto Encyclopedia of Genes and Genomes [KEGG] pathways depict a network of molecular interactions and processes. It contains details on metabolic pathways, signaling pathways and other functional categories [27].

2.6. PPI network analysis with STRING and cytoscape

Protein–protein interaction (PPI) networks were constructed using the STRING database (version 12.0; <http://string-db.org/>) with Homo sapiens as the selected organism [28]. A medium-to-high confidence interaction score threshold of 0.700 was applied, and only experimentally validated and database-sourced interactions were included. Disconnected nodes were removed to improve visualization. The network was imported into Cytoscape software (version 3.10.0; <http://cytoscape.org/>) for further analysis [29]. The CytoHubba plugin was used to identify the hub genes based on multiple topological algorithms, including Degree, Betweenness Centrality, Stress Centrality, Closeness Centrality, EPC, MNC, and MCC [30].

2.7. Construction of target regulatory network – miRNA and TF

Network Analyst Database (www.networkanalyst.ca) was used for predicting target genes and miRNA-TF groups. TarBase v9.0 was used for the miRNA analysis, while chEA3 was used for the TF analysis. Using a degree filter of 6.0, the target regulatory network of the highest relevance has been identified. The miRNA-TF and target gene regulatory network, which exhibits the connections between TFs, miRNAs, and their crucial targets in PCOS, was displayed using Cytoscape [31].

2.8. LASSO regression analysis

Least Absolute Shrinkage and Selection Operator (LASSO) is a machine learning (ML) method that can improve prediction reliability through integrating regularization with variable selection. The hub genes were validated by doing a LASSO with tenfold cross-validation and $\alpha = 0.05$ using the R “glmnet” function. The lasso model retains superior traits while minimizing the coefficients of less significant ones to zero. In lasso regression, the penalty term severity is controlled by a tuning parameter λ . The lasso regression utilizes regularization, which employs a penalty proportionate to the actual size of the coefficients. It can produce a sparse model and reduce coefficients [32].

2.9. ROC curve analysis of key genes

To evaluate the performance of the LASSO regression model, we utilized Receiver Operating Characteristic (ROC) curve analysis. The ROC curve is a graphical representation of the model's diagnostic capacity that plots the true positive rate (sensitivity) against the false positive rate (1-specificity) at different threshold values. The “pROC” package in R was used to perform the ROC analysis [33]. The area under the curve, or AUC, was used to measure the performance of the prediction model. A higher AUC is indicative of better discrimination.

3. Results

3.1. Data pre-processing & alignment of the reads

The statistical power value of the datasets was determined, exceeding the threshold of 0.8, showing that the dataset size was sufficient to detect the significant differences between the two groups (Supplementary Figure 1). This validation supports robustness of the dataset for identifying biological insights. The sequenced reads were

checked for quality using the FastQC tool. The low-quality reads were trimmed by using a trimmomatic tool. After trimming, a second round of quality assessment found a significant improvement in overall read quality, with the majority of reads acquiring a Phred score of 36 or higher, corresponding to a base call accuracy of 99.9 %. Following a read quality assessment, the Hisat2 aligner was employed to align the reads to the human reference genome hg38. We examined each dataset with an overall alignment of over 85 %. The SAM file was converted to BAM format, which generates FeatureCounts for DEG analysis.

3.2. Identification of DEGs

The generated FeatureCounts matrix is used to perform DEG analysis. The R program “DESeq2” was used for gene analysis with $Padj < 0.05$ and $|\log FC| \geq 1$ threshold. The principal component analysis (PCA) plots show the clustering of samples from the control and PCOS groups based on gene expression profiles. The substantial separation of the control and PCOS groups along these principal components indicates that the two groups have distinctive gene expression patterns, as depicted in Fig. 2 (A, B). The dataset SRP224633 (A) has 1193 DEGs, comprising 574 up-regulated genes and 619 down-regulated genes and SRP353681 (B) has 82 DEGs, including 20 up-regulated and 62 down-regulated genes are given in supplementary table 1. 8 Common DEGs are identified in both the datasets as shown in Fig. 3. All DEGs were represented on the volcano map (Fig. 4 (A, B)). The heatmap was generated using the ComplexHeatmap package of R. The heatmap visualizes the expression patterns of the top 30 DEGs found in this study, which were selected based on their statistical significance, as shown in Fig. 5 (A, B).

3.3. GO and pathway enrichment of DEGs

The top DEGs 1193 [SRP224633] and 82 [SRP353681] are further processed for GO analysis. It revealed DEGs of SRP224633 were highly enriched in BP, including chemotaxis ($Padj$: 1.78E-08), taxis ($Padj$: 1.78E-08) and leukocyte migration ($Padj$: 3.67E-09), in addition to CC indicated that the DEGs were enriched in secretory granule membrane ($Padj$: 8.39E-09), endocytic vesicle ($Padj$: 9.10E-06) and cytoplasmic

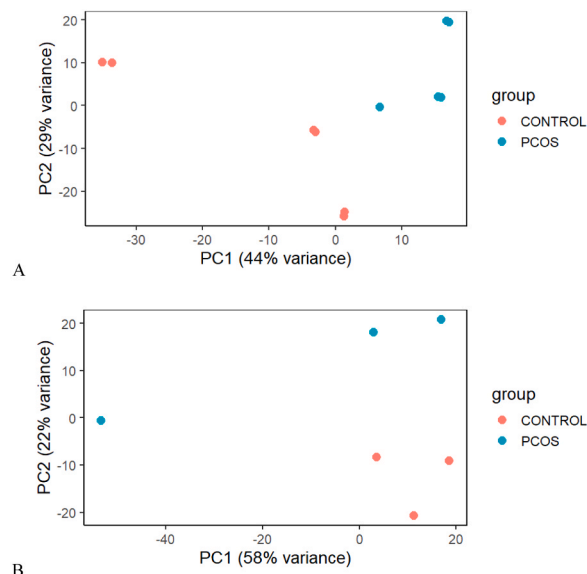


Fig. 2. Principal component analysis plot shows distinct clustering separates PCOS and control samples. In A (SRP224633) and B (SRP353681), indicating distinct gene expression profiles. The x- and y-axes represent the first two principal components explaining most of the total variance. Blue points indicate PCOS samples, and red points indicate control samples.

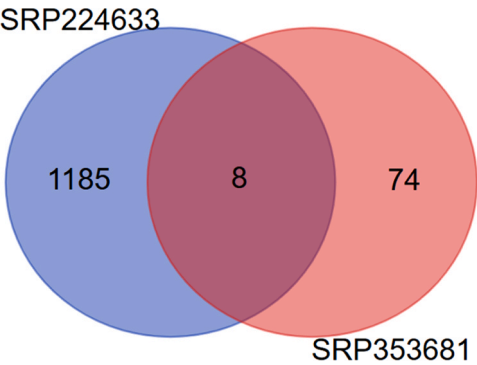


Fig. 3. The Venn diagram shows overlapping differentially expressed genes identified across datasets. It illustrates the number of shared and dataset-specific differentially expressed genes. Overlapping regions represent common DEGs across both datasets.

vesicle lumen (*Padj*: 4.82E-06). In terms of MF, DEGs were highly expressed in nucleoside triphosphatase regulatory activity (*Padj*: 8.39E-05), GTPase regulator activity (*Padj*: 1.76E-05) and organic acid binding (*Padj*: 8.53E-06) which is represented in Fig. 6A. The DEGs of

SRP353681 were highly represented in BP, such as regulation of protein secretion (*Padj*: 0.005218), positive regulation of protein transport, cellular response to lipopolysaccharides (*Padj*: 0.005218). In CC, it was enriched in the endoplasmic reticulum lumen (*Padj*: 0.000336), endocytic vesicle (*Padj*: 0.001395) and transport vesicle membrane (*Padj*: 0.002253163). In addition, MF includes cytokine activity (*Padj*: 0.000202), cytokine receptor binding (*Padj*: 0.000209) and G-protein coupled receptor binding (*Padj*: 0.00066) as shown in Fig. 6B. See [Supplementary Table 2](#) for additional data.

A bubble diagram was generated using the top pathways from the KEGG pathway analysis, collectively with their P-values and the number of genes associated in each pathway. DEGs are highly expressed in SRP224633 pathways such as Chemokine signaling pathway (*Padj*: 9.50E-09), cytokine-cytokine receptor interaction (*Padj*: 2.44E-10) and osteoclast differentiation (*Padj*: 1.32E-13) as well as in SRP353681, highly expressed pathways including the IL-17 signaling pathway (*Padj*: 7.72E-12), steroid hormone biosynthesis (*Padj*: 4.50E-08), viral protein interaction with cytokine & cytokine pathway (*Padj*: 2.18E-12), cytokine-cytokine receptor interaction (*Padj*: 9.89E-14) and chemokine signaling pathway (*Padj*: 2.68E-06) as depicted in Fig. 7 (A, B). Additional results can be found in [Supplementary Table 2](#).

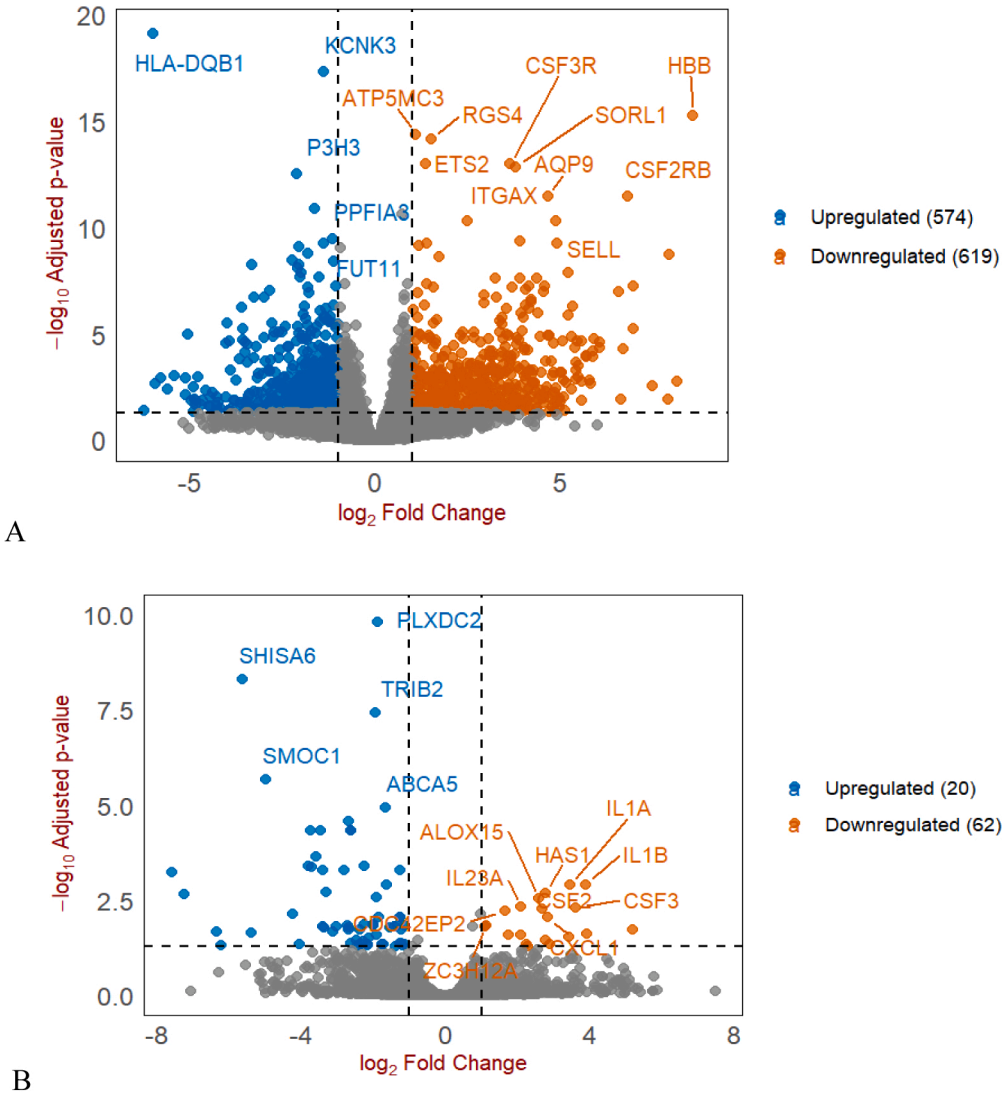


Fig. 4. Volcano plots shows significant gene expression differences genes between PCOS and control groups. In A (SRP224633) and B (SRP353681), $\log_2$ fold change (x-axis) and $-\log_{10}$ adjusted p-value (y-axis) are plotted.

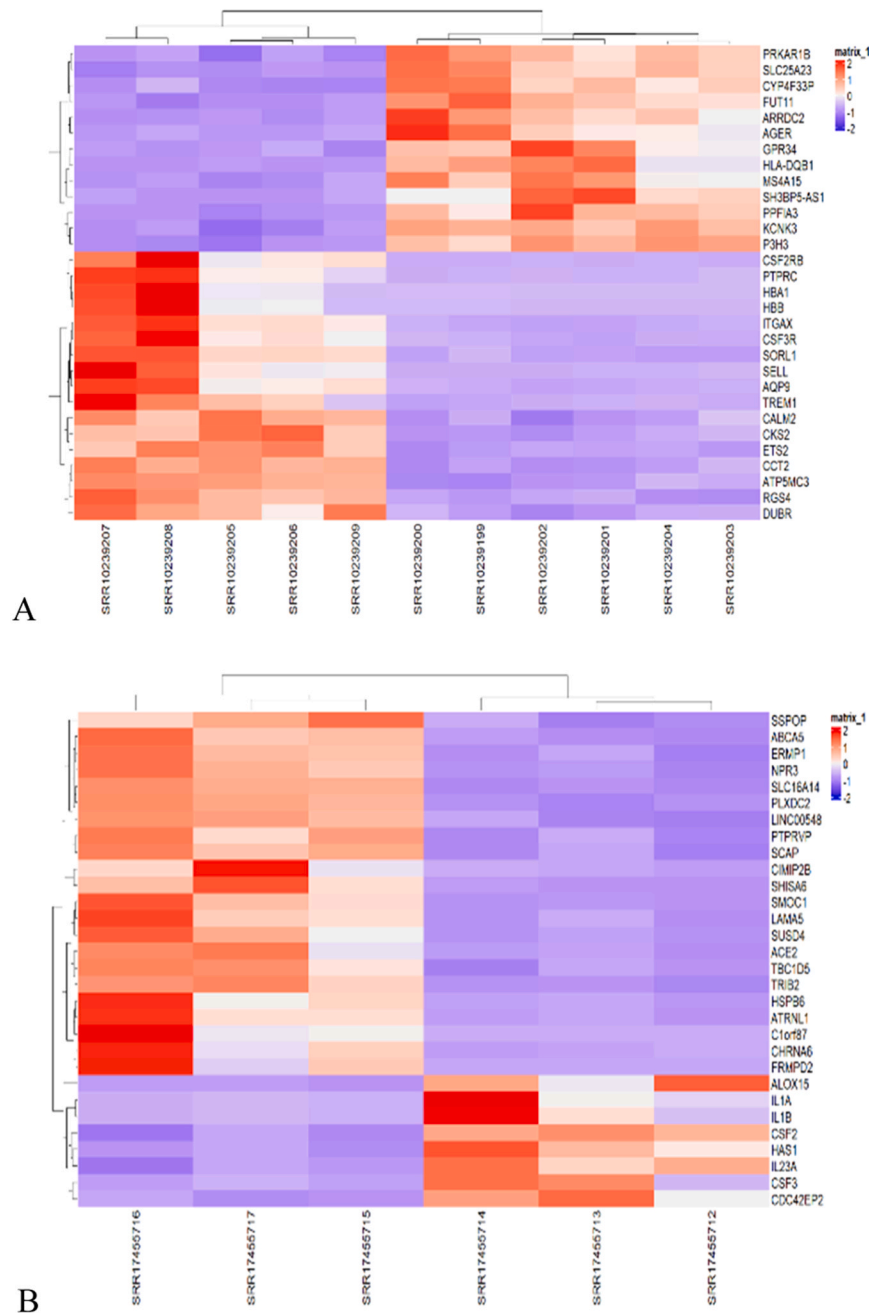


Fig. 5. The heatmap illustrates distinct expression patterns in PCOS and controls. Heatmap from both datasets (A- SRP224633, B- SRP353681) display the expression levels of differentially expressed genes. The color gradient reflects relative gene expression intensities, distinguishing PCOS from control samples.

3.4. Major hub genes in PCOS: PPI network and CytoHubba insights

Following the analysis of the construction of PPI network of DEGs. The significant genes were selected for further analysis. The genes of two datasets, 1267 genes are compiled together to identify the major genes. The PPI is configured to display a maximum confidence score of 0.700 and disconnected nodes are eliminated. It shows 547 nodes and 1847 edges in the STRING database. Hub genes were shown to be associated with PCOS identified in the Cytohubba plugin, including node Degree, Edge Percolated Component [EPC], Maximum Neighborhood Component [MNC], Molecular Complex Detection [MCC], betweenness, stress and closeness and it displayed in Fig. 8. The top hub genes such as Interleukin-1 Beta [IL1B], Protein Tyrosine Phosphatase Receptor Type C [PTPRC], Integrin Subunit Alpha M [ITGAM], TYRO Protein Tyrosine Kinase Binding Protein [TYROBP], Integrin Subunit Beta 2 [ITGB2], Fc

Gamma Receptor IIIa [FCGR3A], C-C Chemokine Receptor 7 [CCR7], Spleen Tyrosine Kinase [SYK], Toll-Like Receptor 2 [TLR2], Fc Gamma Receptor IIIb [FCGR3B], along with their centrality and topology scores of genes are provided in Table 1. Further details are available in Supplementary Table 3.

3.5. Construction of target regulatory network target gene – miRNA and transcription factor

Following the integration of the miRNA and transcription factor (TF) interaction network with the target genes, 24 nodes and 91 edges were discovered. As seen in Fig. 9, key miRNAs and TFs, such as miR-34a-5p, miR-26a-5p, let-7b-5p, Specificity Protein 1 [SP1], Runt-Related Transcription Factor 1 [RUNX1], Estrogen Receptor [ER] and Myelocytomatosis oncogene [MYC], exhibited effective interactions with target

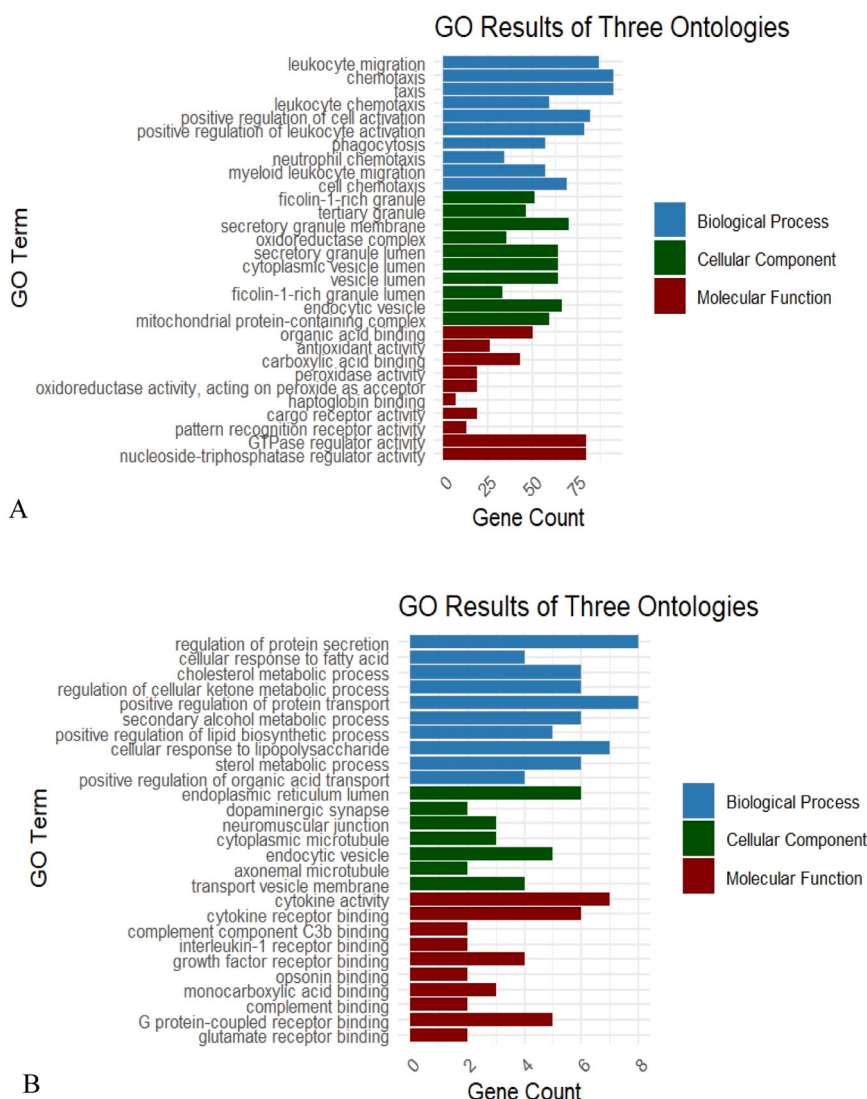


Fig. 6. Gene ontology analysis highlights enriched terms within biological process, cellular component, and molecular function categories. In A (SRP224633) and B (SRP353681), gene counts are plotted on the x-axis, and GO terms on the y-axis. Blue, green, and brown represent BP, CC, and MF, respectively.

genes, including *ITGB2*, *FCGR3A*, *TYROBP*, *SYK*, *FCGR3B*, *ITGAM*, *CCR7*, *IL1B*, and *PTPRC*. *TLR2* was omitted from the network due to its ineffective interactions. The miRNAs such as miR-34a-5p, miR-26a-5p, let-7b-5p, and transcription factors such as *SP1*, *RUNX1*, *ER*, and *MYC* were primarily associated with genes such as *PTPRC*, *IL1B*, *CCR7*, *FCGR3A*, *FCGR3B*, *ITGB2*, *ITGAM*, *TYROBP*, and *SYK*.

3.6. Key gene biomarkers identified by LASSO regression

From the list of identified key genes, which included *ITGB2*, *FCGR3A*, *TYROBP*, *SYK*, *FCGR3B*, *ITGAM*, *CCR7*, *IL1B* and *PTPRC* as identified as potential biomarkers were found using LASSO Poisson regression analysis. Based on lambda min = 1.43457, LASSO regression was performed to identify the most significant variables (Fig. 10A, B). The LASSO model found eleven key genes that exhibited enhanced efficiency. The LASSO model efficiently identified relevant features by reducing coefficients to zero, preventing over fitting as well as rendering to comprehend the possible functions of these genes in PCOS. Due to their highest coefficients in the LASSO model, which indicate their potential significance and pertinent under study, all nine of the key genes can be anticipated as possible biomarkers (Table 2).

3.7. Validation of key gene biomarkers using ROC analysis

The nine key genes identified through LASSO regression—*ITGB2*, *FCGR3A*, *FCGR3B*, *CCR7*, *IL1B*, and *PTPRC*—were subjected to ROC curve analysis to evaluate their diagnostic potential. The individual AUC values for these genes, which are shown in Supplementary Table 4, demonstrated their significant discriminative capabilities. The combined ROC analysis generated a highly desirable AUC value of 0.9 (95 % CI: 0.75–0.95), demonstrating the LASSO model's robustness and reliability of identifying relevant biomarkers.

4. Discussion

PCOS is the most common endocrine condition, characterized by hyperandrogenism and recurrent anovulation [7]. PCOS may significantly impact the quality of life if not treated properly. Undoubtedly, the molecular understanding of considerate syndrome will be helpful in developing their diagnosis and treatment course [9]. Although biomarkers have been discovered as potential treatment targets for PCOS, the exact process of gene regulation that leads to progression of the syndrome is still unknown [8]. The objective of our study was to identify PCOS biomarkers and use bioinformatics analysis to understand their

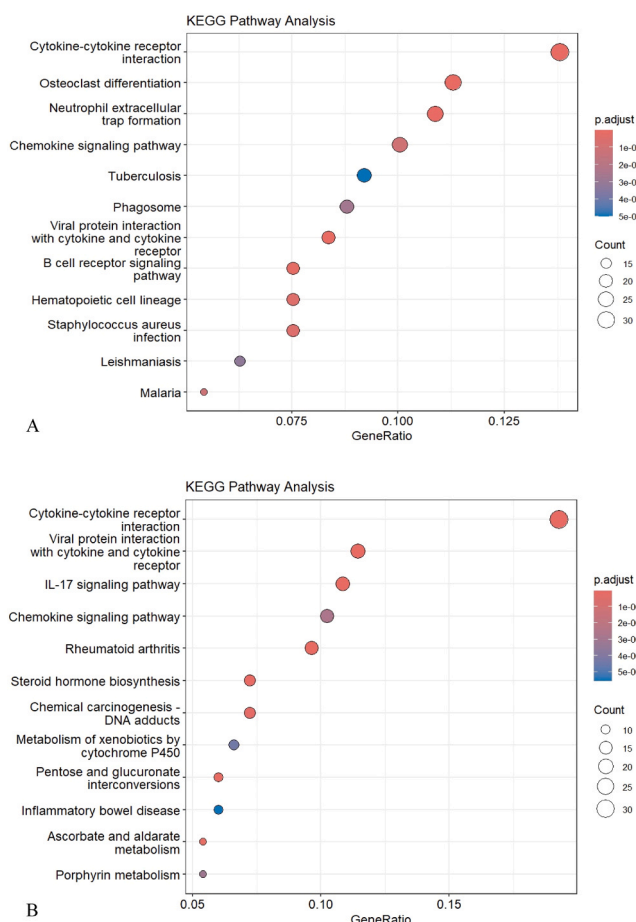


Fig. 7. Bubble plots show pathways significantly enriched in PCOS. In A (SRP224633) and B (SRP353681), x-axis represents the gene ratio, and y-axis lists the pathways. Bubble size indicates the number of associated genes.

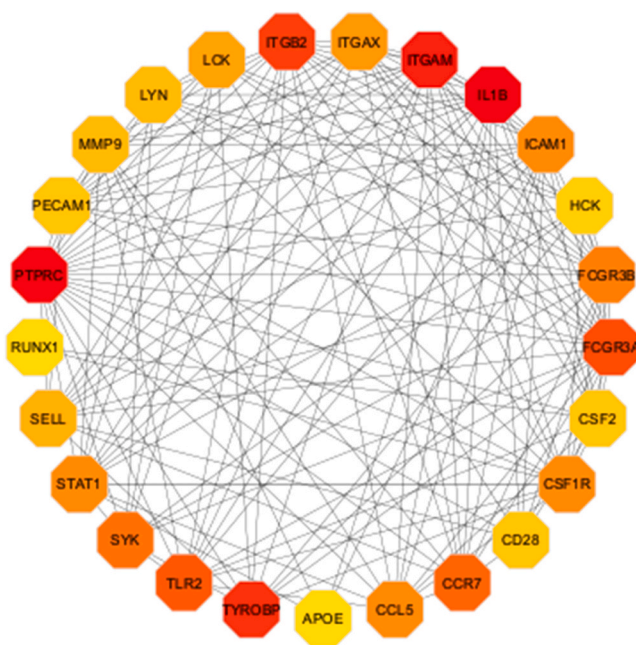


Fig. 8. Network visualization using CytoHubba ranks hub genes by connectivity. Nodes are color-coded from yellow to red, with red indicating the highest-ranked genes.

biological roles. In our investigation, the high throughput sequencing datasets SRP224633 and SRP353681 were used for expression profiling. Consequently, screening of dataset SRP224633 revealed 574 up-regulated and 619 down-regulated genes, while screening of SRP353681 revealed 20 up-regulated and 62 down-regulated genes which are differently expressed between PCOS and normal control samples. GO and KEGG pathway analyzes of DEGs from datasets SRP224633 and SRP353681 provide valuable insights into the molecular mechanisms underlying PCOS. Key DEGs in SRP224633 were significantly enriched in biological processes related to chemotaxis, leukocyte migration and intracellular trafficking, suggesting a possible role in immune response and inflammation. The significant expression of these DEGs in the regulation of nucleoside triphosphatase and GTPase activities points to possible roles in metabolism and cellular signaling. However, DEGs in SRP353681 were highly associated with activities such as regulation of protein secretion and cellular response to lipopolysaccharides, indicating their importance in protein handling and interaction with the immune system. In both datasets, KEGG pathway analysis indicated the importance of inflammatory and immunological signaling pathways such as chemokine signaling, cytokine-cytokine receptor interaction, IL-17 signaling pathway and steroid hormone biosynthesis. These data indicate that chronic inflammation and immunological dysregulation may play significant functions in PCOS pathogenesis [34,35]. Cytokine changes are significant because proper immune system activity is required for a healthy ovulatory process. Immune cells in the ovary secrete local cytokine mediators which promote follicle development, oocyte maturation, follicle rupture on time, angiogenesis, corpus luteum formation, and luteal demise, all of which may be changed in PCOS [36]. The chemokine signaling pathway plays an important part in the pathogenesis of PCOS. Dysregulation of CCR signaling in PCOS is associated with aberrant immune cell recruitment and activation in the ovarian microenvironment, resulting in chronic low-grade inflammation and follicular development abnormalities, both of which are characteristics of the condition [37]. According to research, PCOS is predominantly linked to immune-related pathways and biological reactions in individuals, including neutrophil extracellular formation and T-cell activation [37]. T cells release a variety of inflammatory and immune modulatory substances that assist in the management of ovarian function [38]. Women with PCOS frequently have high levels of chronic inflammation, which might affect leukocyte migration. Changes in molecule adhesion and cytokine levels in PCOS can alter leukocyte migration and function [39]. The analysis results revealed that the key genes with significant variances in expression were mostly observed in cytokine-cytokine receptor interaction, chemokine signaling pathway, leukocyte migration, cell chemotaxis and cytokine activity. A recent study identifies the IL-17 signaling pathway, steroid hormone biosynthesis, malaria and metabolism of xenobiotics by cytochrome P450 as key factors in PCOS development [40], while another identifies the role of osteoclast differentiation, IL-17 signaling and malaria highlight pathways in PCOS pathogenesis [41]. Our study supports these results as these pathways were significantly enriched in our analysis. This shows a significant enrichment of these signaling pathways and highlights their central role in understanding the molecular mechanisms underlying PCOS.

The PPI network and its modules were shown to have a major amount of hub genes linked with PCOS development. The hub genes such as ITGB2, FCGR3A, TYROBP, SYK, FCGR3B, ITGAM, CCR7, IL1B and PTPRC. In a study by Devarbhavi et al., TYROBP and ICAM1 were identified as upregulated in PCOS samples, suggesting their potential involvement in the disease's pathogenesis. Similarly, our study also validates the elevated expression levels of these genes, further strengthening their association with PCOS development [42]. Increased expression of PTPRC is found in the ovaries of women with PCOS. Research has indicated that women with PCOS have altered expression of this gene in granulosa cells and ovarian stroma [43]. The activity of immune cells relies upon the regulation of T and B cell antigen receptor

Table 1
Top 25 key differentially expressed genes obtained by centrality analysis.

GENE	Degree	MCC	DMNC	EPC	Stress	Eigenvector	Betweenness	Closeness
<i>IL1B</i>	69	1987083	0.27148	310.889	284564	0.226223	42952.16	0.369919
<i>PTPRC</i>	60	2768584	0.4089	310.889	169296	0.258189	16971.43	0.359447
<i>ITGAM</i>	55	3433777	0.4267	310.889	119800	0.242467	12926.2	0.353398
<i>TYROBP</i>	50	13725	0.24859	310.889	137116	0.138513	17946.91	0.331713
<i>ITGB2</i>	47	1369356	0.35205	310.889	156758	0.192583	19382.08	0.351351
<i>FCGR3A</i>	44	2351137	0.49309	310.889	79396	0.211351	6279.772	0.33892
<i>TLR2</i>	43	543340	0.36111	310.889	115628	0.160624	16192.37	0.34318
<i>CCR7</i>	39	2180559	0.4826	310.889	59472	0.175819	6307.696	0.32346
<i>SYK</i>	38	13373	0.30122	310.889	86654	0.112382	10548.5	0.325776
<i>FCGR3B</i>	37	2302382	0.50714	310.889	62574	0.188416	4166.681	0.334559
<i>ICAM1</i>	36	2201523	0.49334	310.889	121878	0.175985	15410.51	0.34557
<i>CCL5</i>	36	1113882	0.47253	310.889	38590	0.158098	3640.08	0.324421
<i>STAT1</i>	36	51691	0.28049	310.889	130860	0.147658	11392.36	0.336414
<i>CSF1R</i>	36	27512	0.35949	310.889	99788	0.120601	17162.57	0.344262
<i>ITGAX</i>	34	1920634	0.53033	310.889	55174	0.170041	6024.462	0.329114
<i>LCK</i>	32	23785	0.38191	310.889	63020	0.11575	5820.044	0.317073
<i>SELL</i>	31	2336210	0.5649	310.889	97874	0.154078	11227.77	0.321744
<i>MMP9</i>	30	149442	0.34523	310.889	108880	0.112482	13492.26	0.339552
<i>LYN</i>	30	6625	0.34612	310.889	93888	0.105964	11688.9	0.332927
<i>PECAM1</i>	29	475803	0.43672	310.889	60446	0.136857	2487.529	0.317997
<i>CD28</i>	29	160303	0.46098	310.889	34402	0.135154	6623.518	0.330309
<i>CSF2</i>	29	758312	0.46041	310.889	27366	0.128879	3311.412	0.311822
<i>HCK</i>	28	8404	0.40552	310.889	50160	0.099434	4411.786	0.322314
<i>PLCG2</i>	26	2060	0.36847	310.889	137362	0.120808	4696.086	0.331713
<i>VCAM1</i>	26	259646	0.45604	310.889	54792	0.061853	19883.93	0.322886

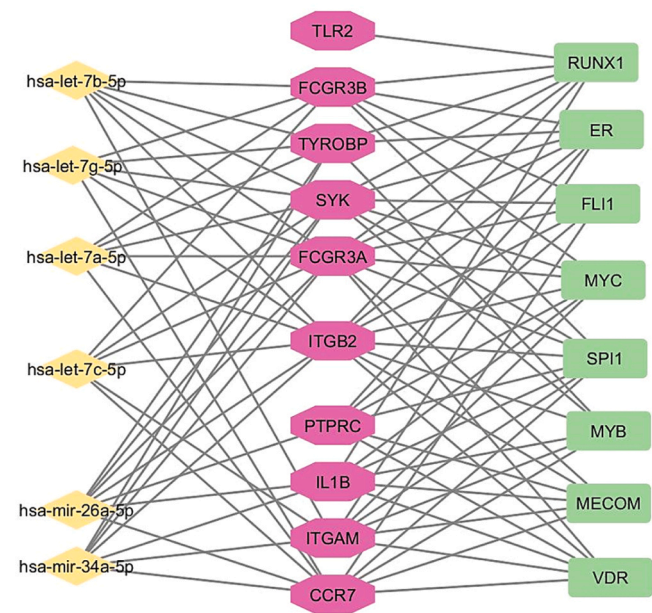


Fig. 9. The network displays the interactions among miRNAs, transcription factors, and key genes related to PCOS. Yellow nodes represent miRNAs, pink nodes represent key genes, and green nodes represent transcription factors.

signaling, largely regulated by *PTPRC*. Its altered expression in PCOS may influence the immune responses and inflammation in the ovaries and other tissues [44]. Several research examined the significance of *IL1B* in PCOS patients because of its stated involvement in inflammatory-related processes in the ovaries. *IL1B* also found to be a common key gene of both datasets. The *ITGB2* gene plays significant functions in cellular adhesion, signal transduction, and immunological responses. *ITGB2* facilitates the infiltration and activation of immune cells by regulating interactions between cells and the extracellular matrix. PCOS is characterized by dysregulated inflammation, and the immunological responses of *ITGB2* may affect follicular growth and the microenvironments of ovarian tissue, aggravating PCOS symptoms. The

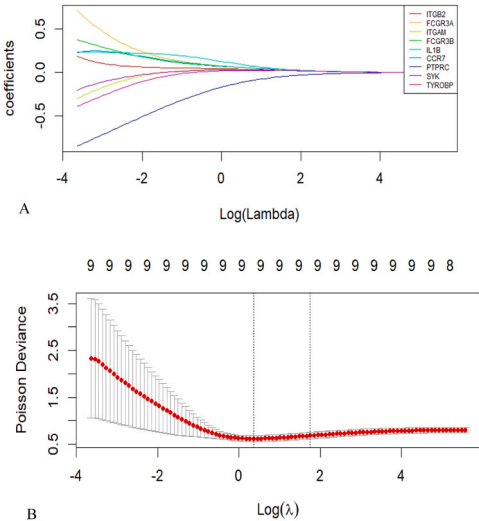


Fig. 10. LASSO regression identifies key predictive genes in PCOS. A) The graph displays the coefficients of the top 9 key genes plotted against the log of the regularization parameter lambda. B) The cross-validation plot for LASSO regression illustrates the link between the regularization parameter (λ) and the Poisson deviation. The Y-axis shows the corresponding Poisson deviation, while the X-axis shows the logarithm of the λ values.

ITGB2 subunit encodes leukocyte-specific integrins, which are crucial for the adhesion and trafficking of immune cells. Amplification of inflammatory signals in ovarian tissue can be mediated by aberrant expression or activity of *ITGB2*, thus affecting the infiltration of immune cells and the local inflammatory environment within the tissue [45]. *ITGB2* has a positive correlation with a higher proportion of immune cells, including CD4 + T cells, CD8 + T cells, B cells, NK cells, and macrophages. *ITGB2* has been found as a prognostic marker for ovarian cancer. Biological function tests demonstrated that *ITGB2* increases the effects of breast cancer's migratory and invasion characteristics [46]. The transcriptomic study identified *ITGB2* as a major hub gene associated with inflammation.

Therefore, *ITGB2* may constitute a significant biomarker due to its

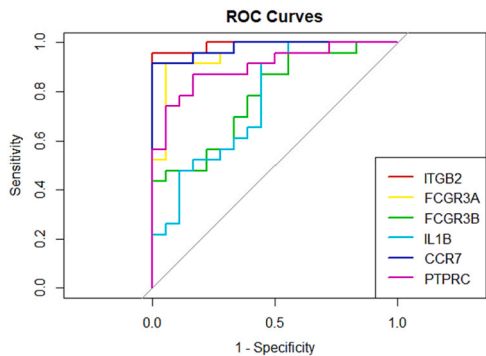


Fig. 11. Receiver operating characteristic curves for six candidate genes (ITGB2, FCGR3A, FCGR3B, IL1B, CCR7, and PTPRC). The curves illustrate sensitivity versus 1-specificity for each gene, with an AUC value of 0.9, demonstrating their strong diagnostic potential.

Table 2
Key genes identified by LASSO regression in PCOS.

S.No	Gene	Coefficient
1	<i>PTPRC</i>	−0.13037
2	<i>IL1B</i>	0.10119
3	<i>CCR7</i>	0.062518
4	<i>FCGR3A</i>	0.062477
5	<i>FCGR3B</i>	0.06075
6	<i>ITGB2</i>	0.041115
7	<i>ITGAM</i>	0.036577
8	<i>SYK</i>	0.030843
9	<i>TYROBP</i>	0.024062

inflammation characteristics in diagnosing PCOS [42]. *FCGR3A* (CD16), a member of the FCGR family, is predominantly expressed on the surface of natural killer (NK) cells, functioning as a transmembrane receptor that enables immune cells to directly target and eliminate harmful cells [47]. *FCGR3A* is highly expressed in various cancer models, including ovarian cancer, and is thought to play a critical role in NK cell-mediated immunity. It was found to be mainly enriched in signaling pathways associated with the negative regulation of angiogenesis, epidermal cell differentiation, chronic inflammation and adaptive immune responses. It is known for its role in activating immune cells by binding to the Fc region of IgG, which can alter inflammatory responses in the ovarian microenvironment by causing chronic low-grade inflammatory conditions, affecting the follicle development and cellular function. However, its role in the development of PCOS is still poorly understood [48]. *FCGR3B*, a neutrophil receptor binds to IgG antibodies, it is essential for clearing immunological complexes and increasing phagocytosis. *FCGR3B* is mainly expressed on neutrophils and contributes to immune responses through phagocytosis and immune complex discharge [49]. *FCGR3B* role in PCOS is not well understood, however it may influence the chronic low-grade inflammation that defines the condition. This inflammation is connected to insulin resistance and ovarian dysfunction, which are important characteristics of the syndrome. Additionally, it has been demonstrated that differentiating sequences, nucleotide polymorphisms, and copy number changes in *FCGR3B* are substantially linked to a number of autoimmune disorders [50]. Aberrant *FCGR3B* expression or polymorphisms have been associated to a variety of autoimmune and inflammatory conditions, increasing the possibility that these conditions may influence the systemic inflammation reported in PCOS. Although direct research associating *FCGR3B* to PCOS remains limited, evidence suggests that immune modulation genes, such as genes in the FCGR family, may be related to disorders characterized by chronic low-grade inflammation, comparable to the inflammatory profile observed in PCOS [51]. Further investigation may provide light on *FCGR3B*'s potential contribution to the inflammatory processes

implicated in PCOS, even though its precise involvement in the disorder is yet unknown. *CCR7*, a chemokine receptor, regulates immune cell migration, particularly similar to dendritic cells (DCs) and T cells, respectively, which are essential for immune response maintenance [52]. The *CCR7* gene has been extensively studied in terms of its role in immune cell trafficking and involvement in immunological-related diseases. According to studies, *CCR7* is required for T cell and dendritic cell migration to lymphoid organs, which influences immunological responses and inflammation. *CCR7* is associated with obesity, as knocking it out in mice protects them against diet-induced obesity. *CCR7* upregulation in adipose tissue indicates adiposity-related inflammation [53]. There is scant but evolving information on the possible connection between immune system dysregulation and PCOS pathogenesis in the context of PCOS. Some research has investigated the larger role of chemokine receptors and inflammatory markers in PCOS, indicating that immune cell activity and chronic inflammation are involved [54]. PCOS patients have been found to have altered immune-related gene expression as well as elevated inflammation. The role it plays in the immune system suggests that it may be relevant to PCOS, which is defined by chronic low-grade inflammation. Chronic inflammation in PCOS may be increased by disruption of the *CCR7-CCL19* [chemokine ligand 19] /*CCL21* [chemokine ligand 21] axis, resulting in prolonged immune cell activation which contributes to tissue damage and metabolic abnormalities [52]. However, specific research investigating *CCR7*'s role in PCOS are sparse.

Analysis of the target gene-miRNA regulatory network and target gene-TF regulatory network could be crucial to understanding PCOS and may potentially result in novel treatment strategies. In PCOS, miR-34a-5p could be involved in follicular growth and oocyte maturation [55]. In PCOS patients, follicular fluid-derived extracellular vesicles have higher levels of miR-34a-5p. It affects granulosa cells by controlling apoptosis and glycolysis via targets such as lactate dehydrogenase A (LDHA) [56]. Inflammatory pathways and insulin resistance are linked to elevated miR-24a-5p levels and these factors are crucial for the metabolic dysregulation recognized in PCOS patients [57]. The insulin signaling pathways that have been linked to miR-7b are essential to the metabolic abnormalities associated with PCOS. Insulin resistance in PCOS patients may be exacerbated by miR-7b's specific association with decreased insulin signaling [58,59]. *SPI1* is also involved in an abnormal miRNA-mRNA regulation network in PCOS. For example, it has been established that there is a negative correlation between *SPI1* expression and the expression of miR-34a, which is associated with inflammation and insulin resistance in PCOS. Increased *SPI1* activity may increase macrophage infiltration and activation, leading to the release of pro-inflammatory cytokines, which may worsen ovarian dysfunction and insulin resistance [60]. *RUNX1* is an important transcription factor in many biological processes, including hematopoiesis, cell differentiation and immunological modulation. Its involvement in several diseases, including cancer and immunological disorders, has been extensively investigated [61]. Although the exact role of *RUNX1* in PCOS is unknown, the inflammatory and ovarian characteristics associated with the disorder may be influenced by its involvement in immune function and gene regulation. Estrogen Receptor (ER) factors play a key role in mediating the effects of estrogen in a range of organs, including the reproductive system. ER are nuclear hormone receptors that regulates the gene expression in response to estrogen binding. They are essential for standard reproductive function, including ovulation and regulation of the menstrual cycle [62]. In the context of our investigation, the miRNAs miR-34a-5p, miR-26a-5p and let-7b-5p, as well as TFs such as *RUNX1*, ER, SP1 and MYC, play critical roles in the regulation of the identified key genes. These miRNA-TF interactions are linked to all of the key genes in our analysis, including both novel genes (*FCGR3A*, *FCGR3B*, *ITGB2* and *CCR7*) and established genes (*PTPRC* and *IL1B*) in PCOS. These miRNAs and TFs act as important regulators in the gene network and regulate the expression and activity of key genes involved in PCOS pathogenesis. Their interaction reveals the complicated

regulatory landscape of the disease and identifies potential areas for therapeutic intervention and further investigation.

Nine key genes have been successfully discovered with high potential as PCOS biomarkers by the use of LASSO Poisson regression analysis; among them, *ITGB2*, *FCGR3A*, *FCGR3B*, *CCR7*, *IL1B* and *PTPRC* are particularly significant. LASSO analysis improves the reliability of identifying key biomarkers by focusing on the most relevant genes and moderating for complexity. The analysis of the ROC curve, which yielded an AUC of 0.9, was performed to validate the significance of the nine key biomarkers identified using LASSO regression analysis, thereby demonstrating strong diagnostic potential. These results confirm the reliability of the genes identified by LASSO and ROC analysis, emphasizing the need for further investigation. To determine the therapeutic value of these biomarkers, future studies should focus on testing them in larger and more diverse populations. In addition, investigating the connections between these genes and other molecular signaling pathways in PCOS could lead to a deeper understanding of the development of the disease.

5. Conclusion

In summary, our investigation revealed that *CCR7*, *FCGR3A*, *FCGR3B* and *ITGB2* are novel PCOS-associated genes that have not been reported previously. These results identify additional genetic factors that may contribute to the pathophysiology of PCOS and open new research opportunities. Furthermore, our results confirm the involvement of *PTPRC* and *IL1B* as predominant genes in PCOS, which is consistent with previous studies. The discovery of these new genes, along with existing ones, illuminates the complicated genetic landscape of PCOS and provides possibilities for future research to investigate these new targets for therapeutic interventions and diagnostic advances.

Limitation

While this study contributes significant insights into potential biomarkers and regulatory pathways of PCOS, recognition needs to be given for the limitations inherent in the datasets used. Our results here are based on RNA-Seq data from SRP224633 and SRP353681, both of which were produced under defined experimental conditions and clinical settings. This could limit the generalizability of our results to heterogeneous ethnic and geographic populations. In addition, the comparatively modest sample sizes within each dataset could restrict the statistical power and stability of the analyses and influence the detection of subtle differential gene expression. Moreover, since this study is based entirely on in silico analyses, the identified genes, pathways, and regulatory networks still require experimental verification of their biological significance. To provide greater translational significance to these results, subsequent studies would benefit from including larger numbers of clinically well-characterized populations from diverse populations and including in vitro or in vivo functional validation.

CRediT authorship contribution statement

A Mohanapriya: Writing – review & editing, Methodology, Conceptualization. **Harshini Senthilkumar:** Writing – original draft, Formal analysis, Data curation, Conceptualization.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Funding

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Declaration of Competing Interest

The authors confirm that they have no personal or financial interests that might bias the objectivity of the research findings.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jsbmb.2025.106887](https://doi.org/10.1016/j.jsbmb.2025.106887).

Data Availability

Data will be made available on request.

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